

3-Mark Questions & Answers -- MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Draw and explain the functional pin diagram of 8086 (any 6-8 important pins).

1. AD0 - AD15 (Pins 2-16, 39): Multiplexed Address/Data lines carrying 16-bit data or lower 16-bit memory address.

2. A1

Q2. Explain memory segmentation in 8086 with a diagram.

```

1 MB Memory Space (00000H to FFFFFH)
+-----+ FFFFFH
|      UNMAPPED MEMORY SPACE      |
+-----+
| EXTRA SEGMENT (ES)      (64 KB) | <- ES:DI
+-----+
| STACK SEGMENT (SS)      (64 KB) | <- SS:SP/BP
+-----+
| DATA SEGMENT (DS)      (64 KB) | <- DS:BX/SI
+-----+
| CODE SEGMENT (CS)       (64 KB) | <- CS:IP
+-----+ 00000H
  
```

- * Total Physical Address Space = 2^{20} = 1 MB.

* Div

Q3. Explain the concept of banking (memory banking) in 8086.

```

1MB PHYSICAL MEMORY
/
512 KB (D0 - D7)      512 KB (D8 - D15)
Connected to A0 = 0    Connected to BHE' = 0
  
```

- * Memory is split into two 512KB banks:

* Ev

Q4. Differentiate between minimum mode and maximum mode operation of 8086.

Feature	Minimum Mode	Maximum Mode
Pin Setup	Pin 33 (MN/MX') connected to +5V.	Pin 33 (MN/MX') connected to GND.
System Architecture	Single CPU processor system.	Multi-processor / Co-processor system
Control Generator	8086 directly outputs `RD`, `WR`, `ALE`	External 8288 Bus Controller decodes s

Q5. Explain the need for demultiplexing the address/data bus in 8086.

1. Pins AD0 - AD15 carry memory addresses during the first clock cycle (T1) and data during T2, T3, T4.

2. W